

Data Collection, Exploration, and Preparation

Data Collection, Exploration, and Preparation – Process View

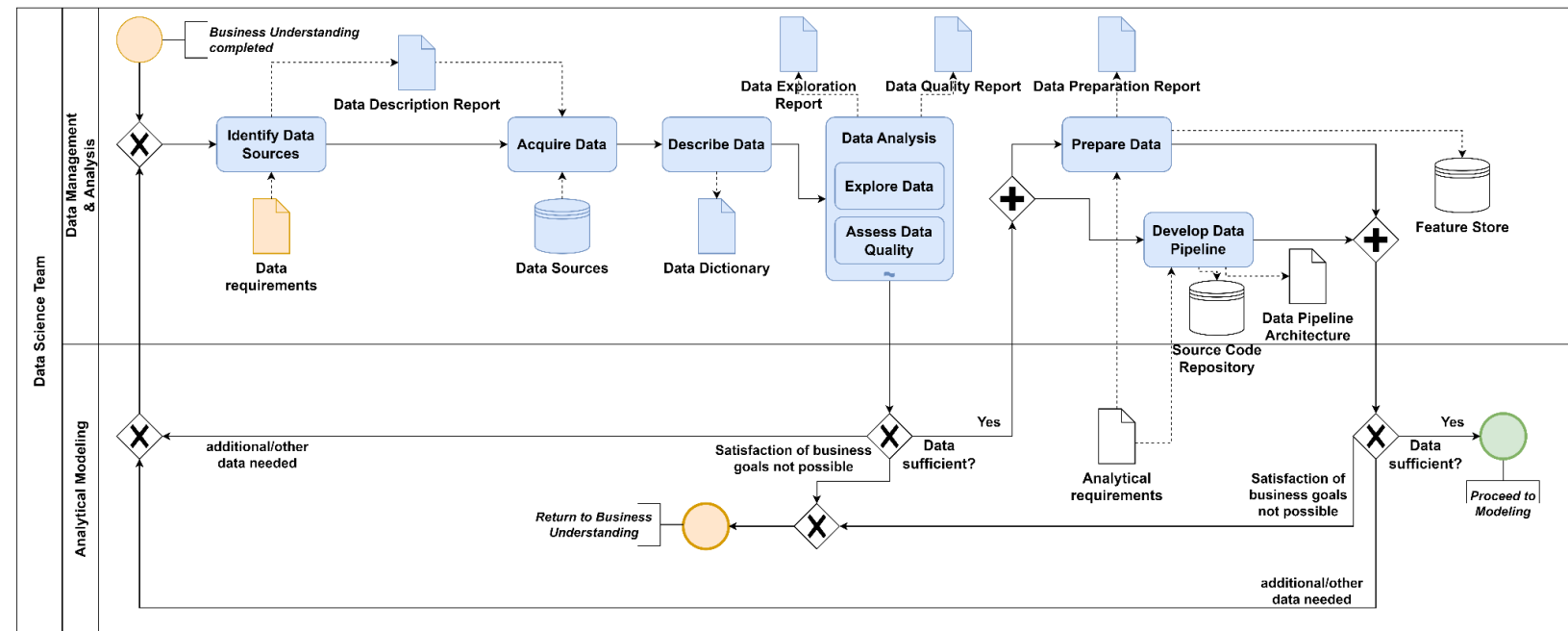
- Identify relevant data sources, initial data analysis, preparation, and feature data pipeline development

- Main tasks

- Identify Data Sources
- Acquire Data
- Describe Data
- Data Analysis
- Prepare Data
- Develop Data Pipeline

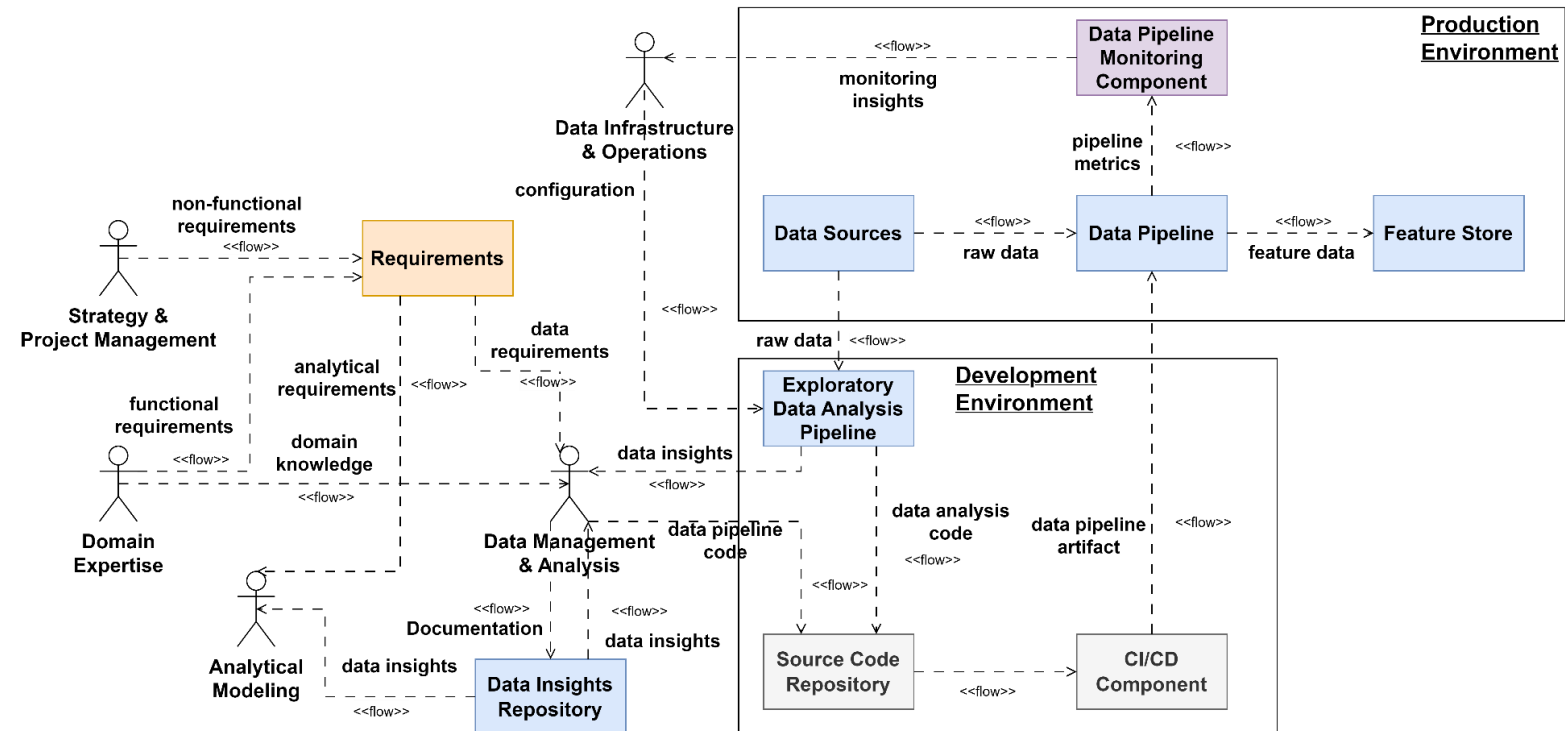
- Artifacts

- Data Description Report
- Data Dictionary
- Data Exploration Report
- Data Quality Report
- Data Preparation Report



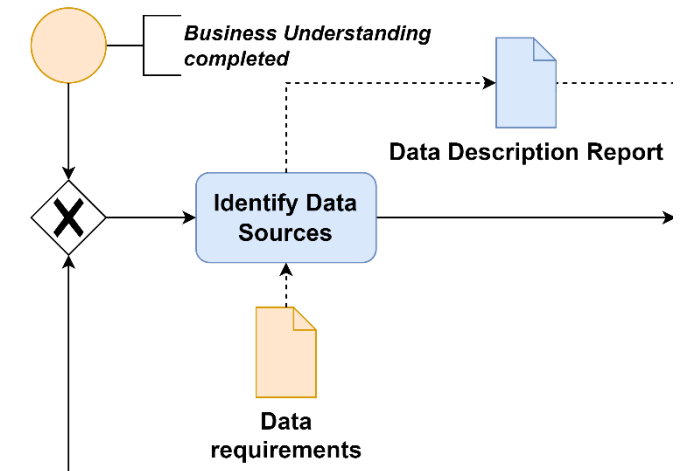
Data Collection, Exploration, and Preparation: Information Flow View

- Separation of environments
- Shaped by data requirements and informed by domain knowledge
- MLOps components
 - source code repository
 - CI/CD component
 - Feature Store
 - Monitoring component (data pipeline)
- Data insights repository: collection of documentation artifacts



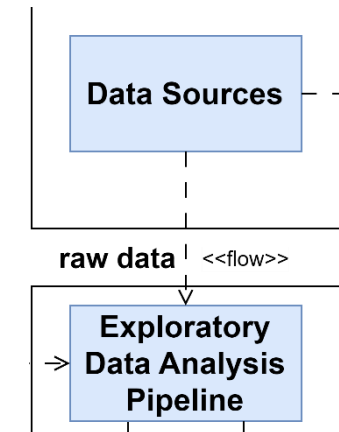
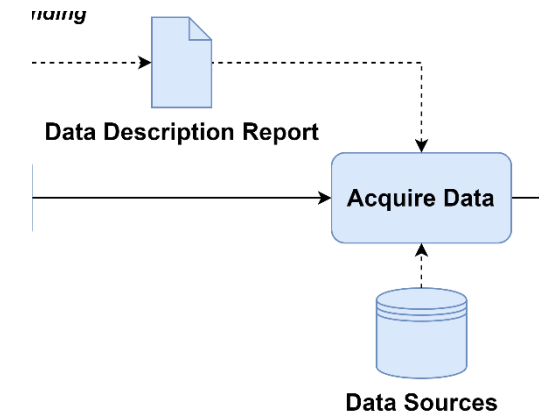
Identify Data Sources

- Goal: identify required and appropriate data sources for the project
- Examples for data sources
 - Files (local/cloud), data warehouses (historical data), online databases (SQL/NoSQL), data streams, online services, incoming messages
 - Can change over time
- Data Description Report
 - Describes involved data sources
 - Name, coverage of data requirements, data extraction methods, effort for acquisition, assumptions for data quality, etc.
 - Lists individual datasets of respective data source that are considered



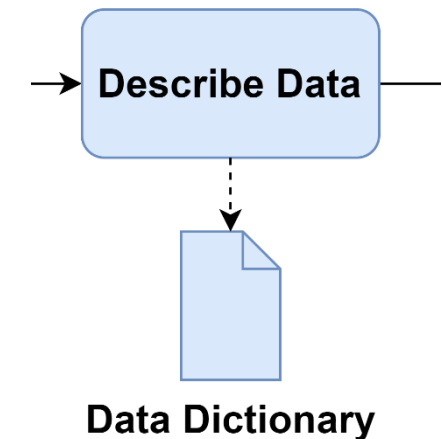
Acquire Data

- Goal: acquire relevant datasets from identified data sources
- Common challenge: data availability
- Prerequisite for data exploration → experimental environment
- Data (subset) is loaded into tool/pipeline used for exploration
 - E.g., notebook



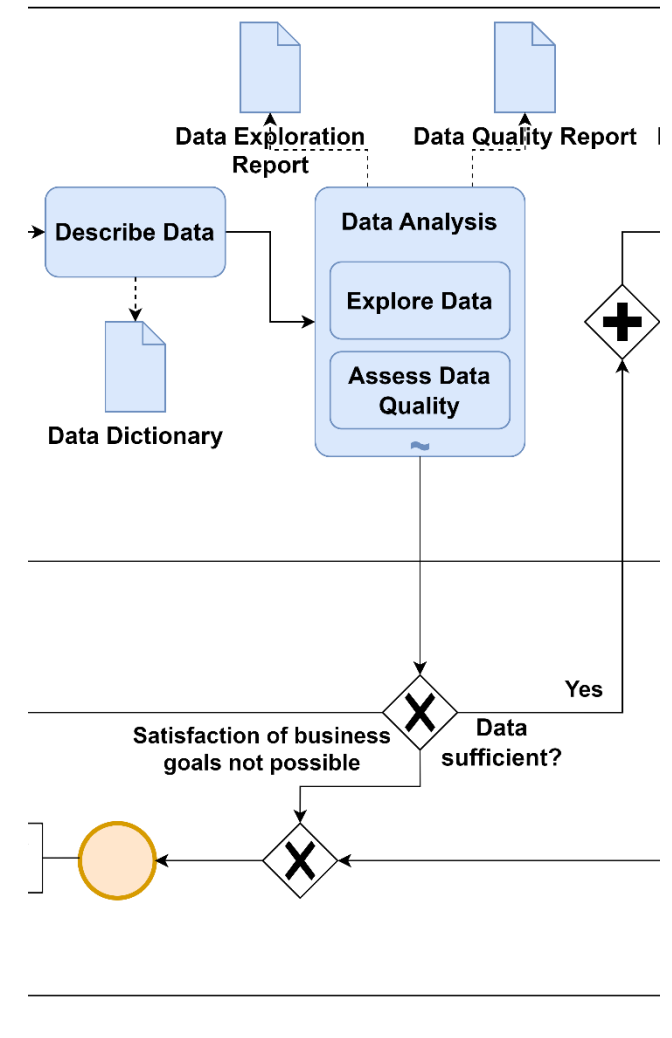
Describe Data

- Goal: precise description of dataset attributes to foster and disseminate data understanding
- Why?
 - Control and governance for data products requires understanding of original data used → data versioning, lineage, and metadata management
 - Factors of ML model reproducibility: code, hyperparameters, and **data**
 - Data versioning is further supported by the feature store (for processed data)
- Data understanding often requires domain expertise
- Data Dictionary
 - Created for each dataset
 - Documents schema, including name, data type, value range, short description, and additional semantic context
 - Can be supplemented by UML class diagrams or ER models



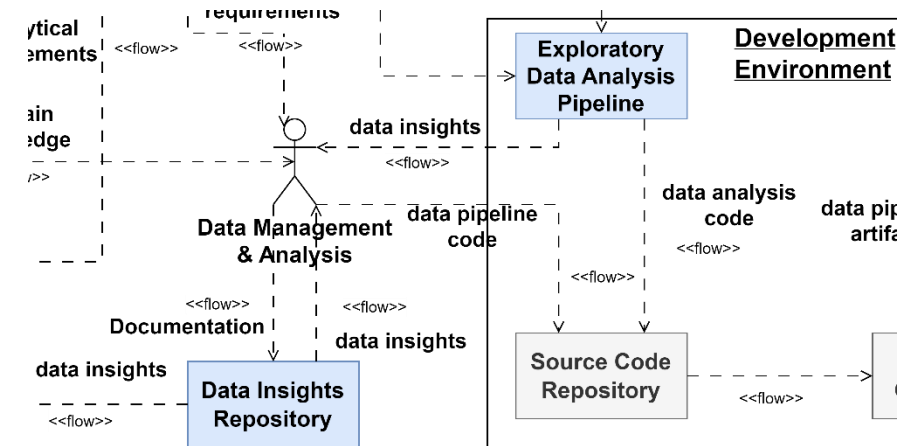
Data Analysis

- Goal: develop a detailed understanding of the data, including flaws in data quality and potential patterns or features for application of analytical models
- deepen preliminary insights from the data description
 - Informs subsequent data preparation and feature engineering
- Dimensions of data quality: accuracy, completeness, consistency, timeliness
- Data Exploration Report
 - Goals and background, applied techniques and activities, results (i.e., semantics, interpretations, implications for data preparation, implications for DS and business goals)
- Data Quality Report
 - Results, approaches and metrics used for assessing data quality, and conclusions for subsequent data preparation



Data Analysis

- Data analysis activities
 - Univariate analysis
 - Summary statistics
 - Distribution visualizations: histograms, box plots, density plots
 - Outlier detection
 - Bivariate/multivariate analysis → relationships
 - Numerical vs. numerical: scatter plots, correlation
 - Categorical vs. numerical: grouped statistics, box plots
 - Categorical vs. categorical: bar charts, contingency table
 - Visualizations are useful to identify trends, clusters, or seasonality
 - Tools: pandas, numpy, matplotlib, seaborn, etc.

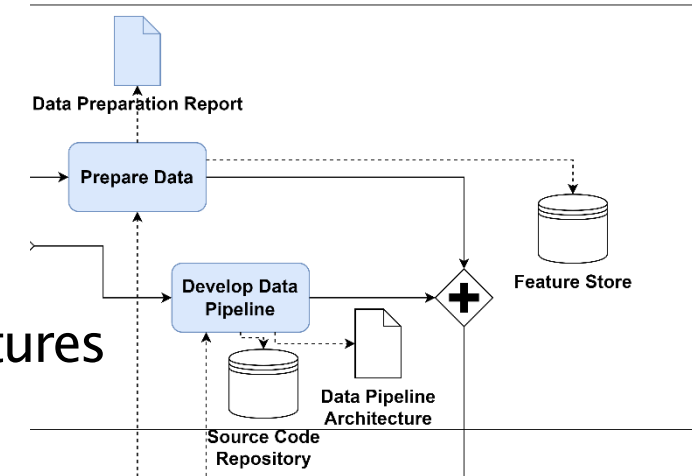


	Group A	Group B	Sum
Likes food	34	67	101
Dislikes food	56	87	143
Sum	90	154	244

Contingency table (example)

Prepare Data

- Goal: construct a target dataset for application of analytical methods
 - Includes data quality fixes and feature engineering
 - Based on the insights from initial data analysis
- Feature engineering
 - Selection, manipulation, and transformation of data to create features used as input for analytical models
- Blueprint for the data pipeline development
- Feature store: central hub for producing, sharing, and monitoring features
- Data Preparation Report
 - Goals and conducted activities, discoveries, and conclusions for further steps
 - Detailed description of each feature (e.g., origin), structure similar to Data Dictionary

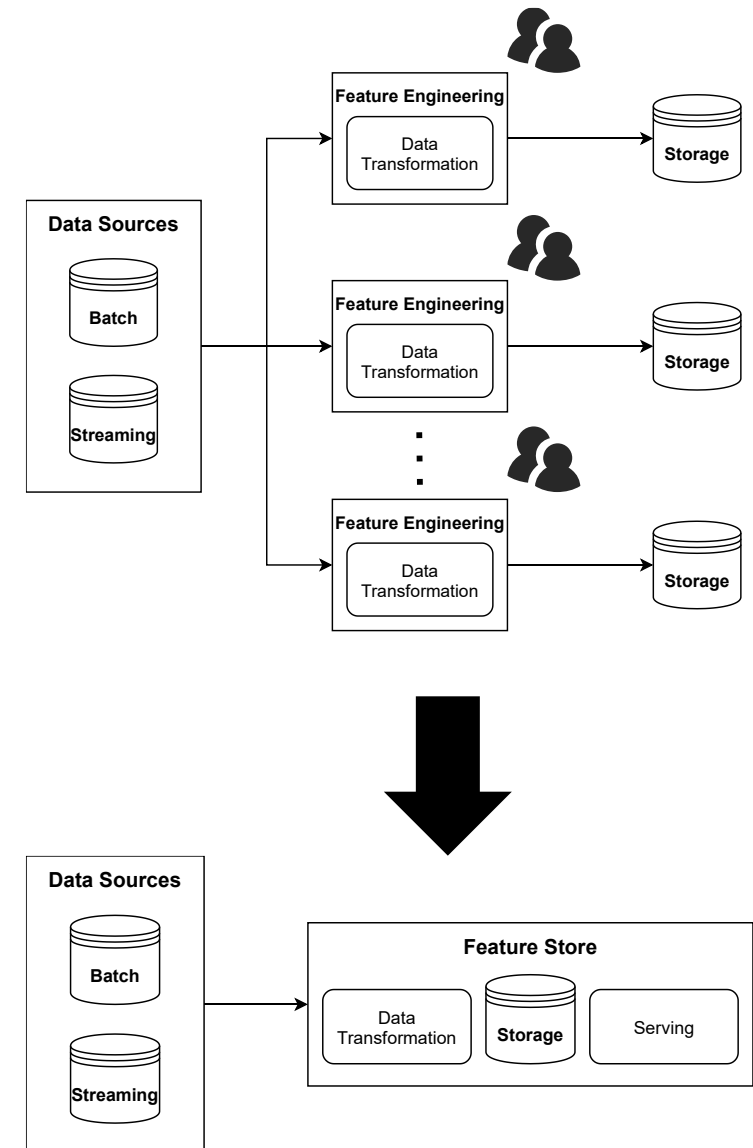


Prepare Data

- **Caution:** data leakage = use of information during model training that would not be available at prediction time
 - Target leakage: predictors include data not available at inference time
 - Train–test contamination: through not distinguishing training from validation data
 - Avoid complex feature engineering that estimates from the dataset before train–test split (performed in Modeling stage)
 - Examples: Mapping/encoding (mapping values from one or more columns to new column values using user–defined operations or code); numerical transformations (log transformation, scaling (e.g., min–max, z–score standardization)), ML methods (clustering, dimensionality reduction, regression, classification, imputing, embedding)
- **Feature Engineering/data preparation activities (examples)**
 - Structured data
 - Filtering: selecting a subset of the data
 - Data integration: combining data from multiple datasets based on common columns
 - Grouping: organizing data into groups
 - Aggregation: calculation of summary statistics
 - Unstructured data
 - Text mining: extract sentiment, entities, and topics from text data
 - Computer vision: image and video processing techniques to extract information from visual data
 - Audio and speech recognition

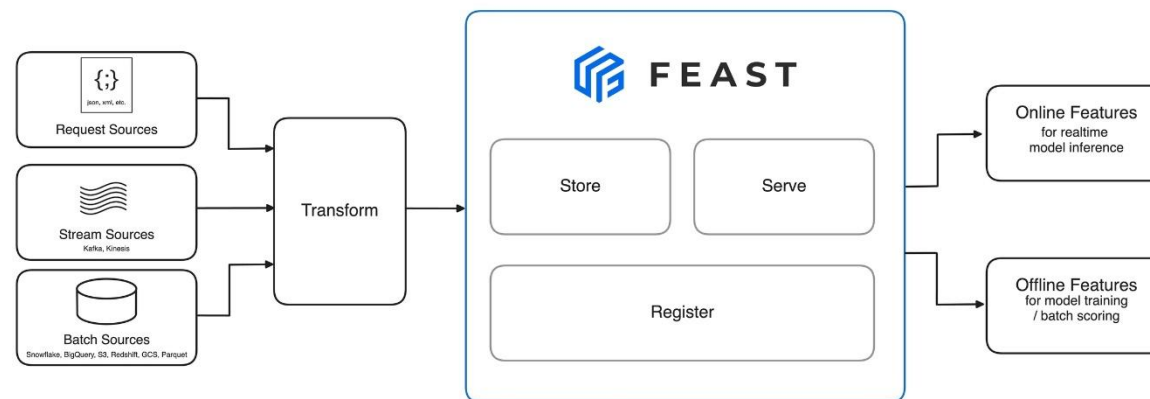
Feature Store

- Centralized feature repository → avoids duplication of work (e.g., multiple definitions for identical features)
- Components
 - Transformation: regularly processing new data into feature values (data pipeline)
 - Storage: offline for training and batch jobs and online for low-latency prediction in production
 - Serving: consistent view of features across training and serving systems
 - Monitoring: detect data drift and training-serving skew
 - Feature registry: central register of standardized feature definitions and metadata
- Considerations
 - Automated pipelines, version control, centralized feature management for consistency and reusability, caching precomputed features, collaboration
- Supplies experimental data pipeline for analysis, experimental ML pipeline, production ML pipeline, and serving component → accelerates experimentation, retraining, and prediction services



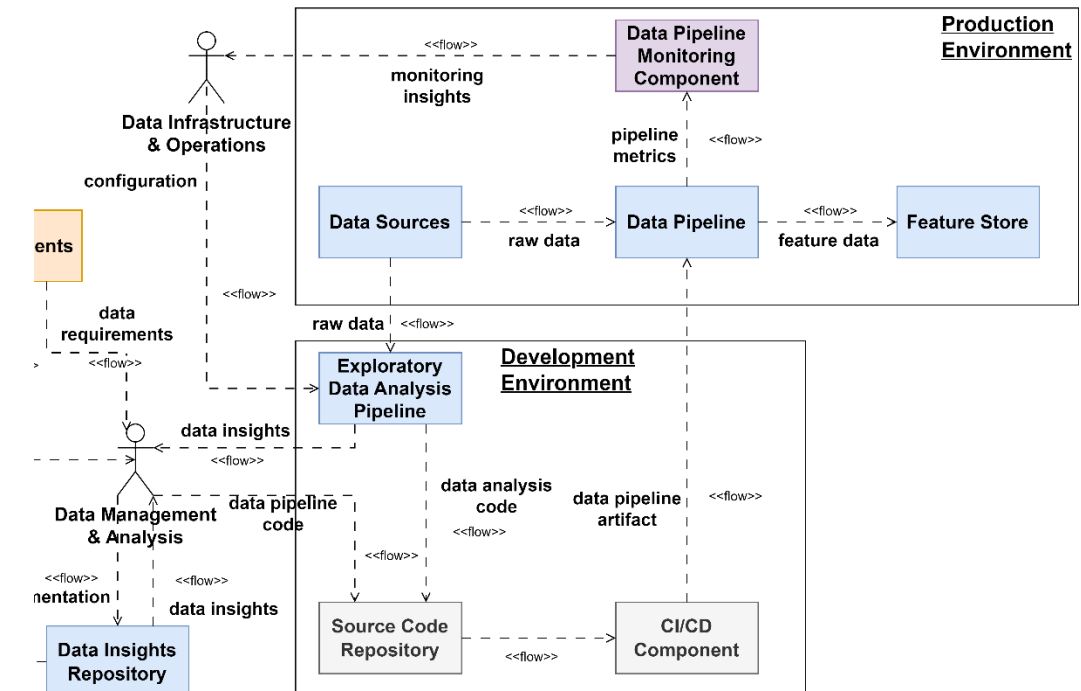
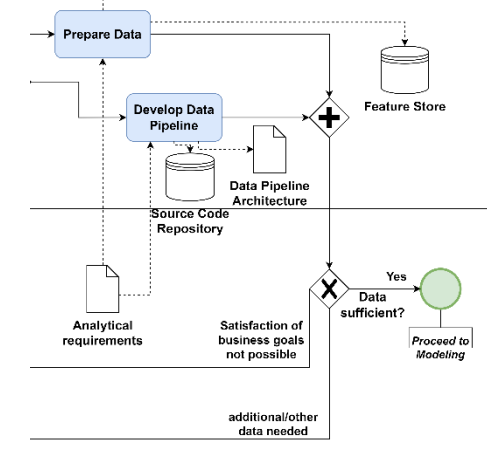
Feature Store

- Tools: Feast, Tecton, MLRun, Sagemaker, Vertex AI, Databricks, Hopsworks
- Feast (<https://docs.feast.dev/>)
 - No transformation service provided → data should be prepared upfront and stored in a supported source
 - Registers source dataset and its metadata (schema, entity, etc.) in a FeatureView object
 - „Materialization“ run to make features available in online store
 - Data in store is inaccurate between materializations
 - Running too many materializations can cause significant computing overhead



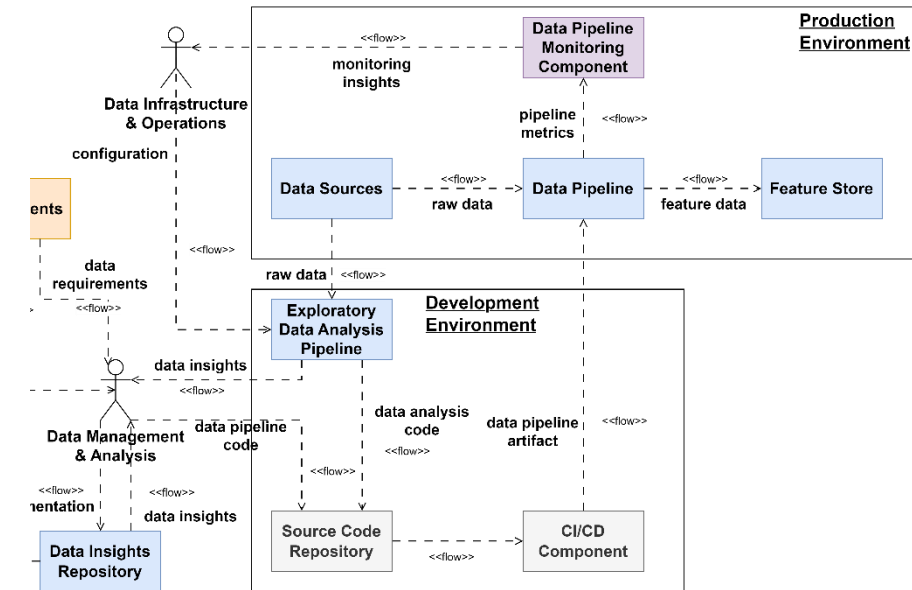
Develop Data Pipeline

- Goal: automation of feature data delivery
 - enables efficient data delivery and model retraining in production → automates data extraction and preparation
- Data pipeline: input from one or more data sources is processed through multiple stages, resulting in a target data set
 - Similar to the data preparation output
 - Output versioned in a feature store
 - Tests and monitoring → quality assurance
- Commonly performed by data engineers
- Data lineage: tracking of data information from creation to consumption
- Data Pipeline Architecture
 - Describes the architecture of the pipeline visually or textually



Source Code Repository and CI/CD (Data Pipeline)

- Purpose: multiple developers can contribute code updates to a shared project repository, conduct automated testing, and have a controlled and continuous deployment process
 - Faster time to market, fewer resources, lower software failure rates
- Versioning extends beyond code and incorporates data source objects, parameters, and multiple artifacts
- Testing and monitoring are more complex in ML systems, include data, infrastructure, model, prediction, system etc.
- CI/CD flow for data pipelines
 - Development: new development branch is created from latest branch, features are added, local tests conducted
 - Staging/integration: request to merge new feature into dev branch → automated test procedures → merge
 - Deployment to production: once new version is verified to work correctly → release (rollback possible)

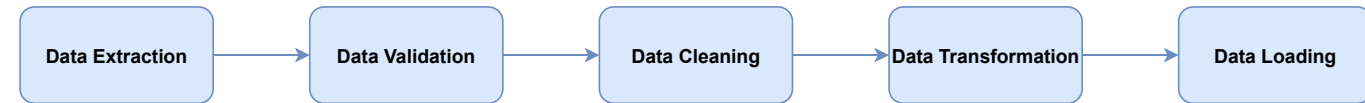


Data Pipeline: Testing and Monitoring

- Data testing:
 - Unit tests: independent units within a pipeline step behave as expected
 - Integration tests: interaction of pipeline steps work appropriately
 - Smoke tests: basic checks to confirm plausibility of pipeline (intermediate) outputs
 - Load tests: testing pipeline under heavy conditions
 - ...

- Data Pipeline Monitoring

- data
 - Schema tests
 - Data quality tracking: range checks, constraints, data formats, etc.
 - Data drift detection: stability of statistical properties
 - Feature expectations: match with training expectations (e.g., new categories)
 - Avoid use of unlawful/private data
 - ...
 - System and infrastructure
- Tools: Great Expectations, Evidently AI



Exemplary data pipeline

References

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